NOIDA INSTITUTE OF ENGINEERING & TECHNOLOGY (NIET), GREATER NOIDA
Department of Computer Science & Engineering (Artificial Intelligence)

SMART ATTENDANCE MANAGEMENT SYSTEM

Complete Project Explanation, Line-by-Line Code Breakdown & Viva Prep (Easy Hinglish Guide)

B.Tech 2nd Year (3rd Sem) | Capstone Project / PBL | Group 46 | Faculty Mentor: Dr. Sarika Agrawal

Univ Roll No	Student Name	Assigned Module	Java Class Handled
2201331520045	**Krish Kumar (Team Lead)**	Business Logic & 75% Formula	Student.java, AttendanceManager.java
2201331520015	**Ashutosh Jitendra Singh**	Data Persistence & CSV Engine	FileManager.java, students.csv
2201331520058	**Suman Kumar Gorain**	Swing GUI & Batch Checkbox Table	SmartAttendanceUI.java
2201331520052	**Preety Kumari**	Input Validation & QA Matrix	ValidationHelper.java
2201331520038	**Govind Kumar**	Architecture & OOP Relationships	System Flow & Documentation

SECTION 1: SABSE PEHLE SAMAJHO — PROJECT ME HO KYA RAHA HAI? (BIG PICTURE)

Bhai simple words me flow samjho:
1. **App Launch:** Jaise hi project run hota hai, FileManager sabse pehle data/students.csv file ko check karta hai. Agar file pehle se hai toh wahan se students ka pura data (Roll, Name, Total Classes, Attended Classes) memory (ArrayList) me load kar leta hai. Agar file nahi mili, toh crash hone ke bajaye turant khud ek nayi file bana deta hai default 8 students ke sath (taaki Mam ke samne error na aaye).
2. **Main Menu:** Screen khulti hai jisme teen simple buttons hain: **Teacher Login**, **Student Portal**, aur **Exit**.
3. **Teacher Flow:** Teacher Login dabane par password mangta hai (admin123). Password sahi hone par **Faculty Dashboard** khulta hai. Yahan se teacher 4 main kaam kar sakta hai:
• **Mark Batch Attendance:** Ek single window table khulti hai jisme saare students ke aage Checkboxes lage hain. Teacher ek click me 'Select All' ya 'Deselect All' kar sakta hai, aur 'Submit Attendance' dabane par sabki total classes +1 ho jaati hain aur jo present the unki attended classes +1 ho jaati hain. Ye data turant CSV file me permanently save ho jata hai!
• **Add New Student:** Naya student add karta hai. Agar teacher negative number (-5), zero, ya pehle se registered roll number daalta hai, toh system usko warn karke rok deta hai.
• **View All Reports:** Sabhi students ki attendance list percentage aur status ke sath table me dikhti hai.
• **Defaulter List (< 75%):** Ek click me sirf unhi bacho ko filter kar deta hai jinki attendance 75% se kam hai.
4. **Student Flow:** Student Portal me student apna Roll No daalta hai. System uska digital card dikhata hai — percentage, Green me **ELIGIBLE FOR EXAM** ya Red me **CRITICAL / SHORT ATTENDANCE**, aur sath me advice deta hai ki 75% reach karne ke liye aage kitni consecutive classes attend karni padengi ya kitni classes bunk kar sakte hain safe zone me.

SECTION 2: TEAM LEADER OPENING SPEECH (MAM KE SAMNE KYA BOLNA HAI)

Krish Kumar (Team Lead) ko stage start karte waqt ye word-to-word bolna hai. Isko normal confidence se bolna:

"Good morning Ma'am. We are Group 46, and our Capstone Project is the **Smart Attendance Management System with CSV Persistence & Bulk Marking**.

Ma'am, pehle wale prototype me 2 major problems thi:
1. Data sirf RAM me tha, jaise hi application band hoti thi, saare newly added students aur attendance records delete ho jaate the.
2. Attendance lagane ke liye 60 bacho ke liye 60 pop-up boxes aate the, jisme teacher ko bar-bar Yes/No click karna padta tha jo practical nahi tha.

Review 2 me humne pure project ko 4-tier modular architecture me re-design kiya hai:
• Data Persistence ke liye humne **CSV File Engine** banaya hai jo har session me data save rakhta hai.
• Attendance ke liye humne **Single-Screen Batch Table with native JCheckBoxes** banaya hai jisse 30 seconds me pure batch ki attendance lag jaati hai.
• **Input Validation Layer** banaya hai jo duplicate roll numbers aur invalid inputs ko block karta hai.
• Aur institute ke **75% attendance rule** ke hisaab se automated Defaulter Filter aur Student Advisory banaya hai.

Ab humare team members apne-apne assigned modules ko practically show karenge."

SECTION 3: CLASS-BY-CLASS & LINE-BY-LINE COMPLETE CODE BREAKDOWN

Agar Mam code khol kar kisi bhi line ya method par ungli rakh kar puchhein, toh yahan se dekh kar bilkul simple Hinglish me samjha dena:

1. Student.java — Model Layer (Lead: Krish Kumar)

Ye class kya karti hai? Ye ek POJO (Plain Old Java Object) hai jo har individual student ka data (Roll, Name, Total Classes, Attended Classes) aur percentage ka formula store karti hai.

```
private int rollNo;
private String name;
private int totalClasses;
private int attendedClasses;
```

Line Meaning: Ye chaar private variables hain. **Kyu likha:** OOPs ka Encapsulation follow karne ke liye variables ko private rakha gaya hai taaki bahar ki koi class inko directly galat modify na kar sake. Inko access karne ke liye public getters/setters hain.

```
public Student(int rollNo, String name) {
    this(rollNo, name, 0, 0);
}
```

Line Meaning: Ye constructor tab call hota hai jab naya student register hota hai. Naye student ki total aur attended classes shuru me 0 hoti hain. this(rollNo, name, 0, 0) likh kar humne constructor chaining ki hai jo 4-parameter wale constructor ko call karta hai.

```
public Student(int rollNo, String name, int totalClasses, int attendedClasses) {
    this.rollNo = rollNo;
    this.name = name;
    this.totalClasses = totalClasses;
    this.attendedClasses = attendedClasses;
}
```

Line Meaning (Constructor Overloading): Ye constructor tab use hota hai jab hum CSV file se purana attendance data reload karte hain. Isse purane attendance counters (jaise 24 me se 22 classes) wapas restore ho jaate hain.

```
public void markAttendance(boolean isPresent) {
    this.totalClasses++;
    if (isPresent) {
        this.attendedClasses++;
    }
}
```

Line Meaning: Jab attendance mark hoti hai, toh class count hamesha 1 badhti hai (totalClasses++). Agar student present tha (isPresent == true), toh uski attendedClasses bhi +1 ho jaati hai. Agar absent tha, toh sirf total badhti hai.

```
public double getPercentage() {
    if (totalClasses == 0) return 0.0;
    return ((double) attendedClasses / totalClasses) * 100.0;
}
```

Line Meaning: Formula: (attended / total) * 100. Yahan (double) me typecast isliye kiya kyunki agar dono integer rahenge toh Java me 22/24 ka answer 0 ban jayega! Double karne se accurate 91.67% aata hai. Aur agar totalClasses 0 ho, toh 0.0 return karte hain taaki Divide By Zero error na aaye.

```
public boolean isDefaulter() {
    return getPercentage() < 75.0;
}
```

Line Meaning: Check karta hai ki student ki attendance 75% se kam hai ya nahi. Agar kam hai toh true return karega.

2. FileManager.java — Data Persistence Layer (Lead: Ashutosh Jitendra Singh)

Ye class kya karti hai? Ye file handling ka engine hai jo data ko hard disk par data/students.csv file me read aur write karti hai, taaki app band hone ke baad bhi data bacha rahe.

```
private static final String FILE_PATH = "data/students.csv";
```

Line Meaning: Relative path define kiya hai jahan file store hoti hai. static final isliye hai taaki ye constant rahe.

```
public static ArrayList<Student> loadStudents() {
    ArrayList<Student> list = new ArrayList<>();
    File file = new File(FILE_PATH);
    if (!file.exists()) { createDefaultFile(file); }
    try (BufferedReader br = new BufferedReader(new FileReader(file))) { ... }
}
```

Line Meaning:
• if (!file.exists()): Agar file pehle se disk par nahi hai, toh crash hone ki jagah createDefaultFile() call karta hai jo turant sample students ke sath nayi file create kar deta hai.
• BufferedReader br: FileReader ke upar BufferedReader lagaya hai kyunki BufferedReader me 8KB ka memory buffer hota hai, jo file ko fast line-by-line read karta hai.
• try (...): Ye Java 7/8 ka **Try-with-resources** hai. Iska fayda ye hai ki file padhne ke baad stream automatically close ho jaati hai, hume alag se br.close() likhne ki jarurat nahi hoti.

```
while ((line = br.readLine()) != null) {
    if (line.isEmpty() || line.toLowerCase().startsWith("rollno")) continue;
    String[] parts = line.split(",");
    int roll = Integer.parseInt(parts[0].trim());
    String name = parts[1].trim();
    int total = Integer.parseInt(parts[2].trim());
    int attended = Integer.parseInt(parts[3].trim());
    list.add(new Student(roll, name, total, attended));
}
```

Line Meaning: Jab tak file me lines hain, ek-ek line padhta hai. Header row ('RollNo,Name,...') aur empty lines ko skip kar deta hai. Comma (,) ke hisaab se tukde karta hai (split), String ko number me convert karta hai (Integer.parseInt), aur naya Student object banakar list me jod deta hai.

```
public static boolean saveStudents(ArrayList<Student> list) {
    try (PrintWriter pw = new PrintWriter(new FileWriter(file))) {
        pw.println("RollNo,Name,TotalClasses,AttendedClasses");
        for (Student s : list) {
            pw.println(s.getRollNo() + "," + s.getName() + "," + s.getTotalClasses() + "," +
s.getAttendedClasses());
        }
        pw.flush();
        return true;
    }
}
```

Line Meaning: Ye memory me rakhi list ko wapas file me likhta hai. pw.println se header aur records comma-separated likhe jaate hain. pw.flush() sabse important hai — ye Java ke internal buffer ko force karta hai ki byte data ko turant physical hard disk par write kare taaki achanak power cut ya crash hone par bhi file corrupt na ho.

3. ValidationHelper.java — Input Validation Layer (Lead: Preety Kumari)

Ye class kya karti hai? Ye security aur boundary check karti hai taaki koi galat, negative, khali ya duplicate roll number na daal sake.

```
public static boolean isPositiveInteger(String text) {
    if (text == null) return false;
    try {
        int roll = Integer.parseInt(text.trim());
        return roll > 0;
    } catch (NumberFormatException e) {
        return false;
    }
}
```

Line Meaning: Ye method check karta hai ki roll number sahi positive integer hai ya nahi. Agar user ne 'ABC' ya '-5' likha: 'ABC' likhne par NumberFormatException aayegi jo catch me pakad kar false return karegi (app crash nahi hogi!). Aur '-5' ya '0' likhne par roll > 0 false ho jayega. Sirf 1, 2, 101 jaise valid roll numbers hi accept honge.

```
public static boolean isRollDuplicate(ArrayList<Student> list, int roll) {
    if (list == null) return false;
    for (Student s : list) {
        if (s.getRollNo() == roll) return true;
    }
    return false;
}
```

Line Meaning: Duplicate check loop! Naya student add karne se pehle ye loop pure studentList ko scan karta hai. Agar wahi roll number pehle se kisi student ka hai, toh true (duplicate) return kar deta hai taaki teacher galti se ek hi roll number do baar register na kar de.

```
public static boolean isNonEmpty(String text) {
    return text != null && !text.trim().isEmpty();
}
```

Line Meaning: Check karta hai ki student ka naam null na ho aur na hi sirf spaces (empty) ho.

4. AttendanceManager.java — Business Logic Controller (Lead: Krish Kumar)

Ye class kya karti hai? Ye GUI aur File ke beech ka Controller hai. UI directly file se baat nahi karti, UI is class ko bolti hai, aur ye class data update karke file ko bolti hai.

```
public AttendanceManager() {
    this.studentList = FileManager.loadStudents();
}
```

Line Meaning: Constructor me hi FileManager.loadStudents() ko call karke master studentList tayyar kar li jaati hai.

```
public boolean markBatchAttendance(boolean[] attendanceArray) {
    for (int i = 0; i < studentList.size(); i++) {
        studentList.get(i).markAttendance(attendanceArray[i]);
    }
    return FileManager.saveStudents(studentList);
}
```

Line Meaning (Batch Algorithm): Teacher ne table me jo-jo checkboxes tick ya untick kiye the, unka boolean array pass hota hai. Loop chalakar har student ka markAttendance() call hota hai aur turant FileManager.saveStudents() se file update ho jaati hai.

```
public ArrayList<Student> getDefaulters() {
    ArrayList<Student> defaulters = new ArrayList<>();
    for (Student s : studentList) {
        if (s.isDefaulter()) { defaulters.add(s); }
    }
    return defaulters;
}
```

Line Meaning: Filter logic! Ye list me se sirf unhi students ko nikal kar alag list banata hai jinka isDefaulter() == true (< 75%) hai.

5. SmartAttendanceUI.java — Swing GUI Presentation Layer (Lead: Suman Kumar Gorain)

Ye class kya karti hai? Ye pura visual graphical interface manage karti hai — Dashboard, Buttons, Colors, JTable with Checkboxes, aur Pop-up Dialogs.

```
DefaultTableModel model = new DefaultTableModel(columnNames, 0) {
    @Override
    public Class<?> getColumnClass(int columnIndex) {
        if (columnIndex == 4) return Boolean.class;
        return Object.class;
    }
    @Override
    public boolean isCellEditable(int row, int column) {
        return column == 4;
    }
};
```

Line Meaning (Sabse Important Swing Trick!):
• **getColumnClass(4) -> Boolean.class:** Java Swing me JTable ko jab hum bolte hain ki Column 4 ka type Boolean.class hai, toh Swing bina kisi third-party library ke khud-ba-khud wahan clickable interactive **JCheckBox** render kar deta hai!
• **isCellEditable() -> column == 4:** Humne sirf column 4 ko editable rakha hai. Isse teacher Roll No ya Student Name ko galti se edit nahi kar sakta, sirf checkbox ko tick/untick kar sakta hai.

```
selectAllBtn.addActionListener(e -> {
    for (int i = 0; i < model.getRowCount(); i++) model.setValueAt(Boolean.TRUE, i, 4);
});
```

Line Meaning: 'Select All' button dabate hi loop chalta hai aur table ke har row ke column 4 me Boolean.TRUE set kar deta hai. Sabhi students ek sath Present mark ho jaate hain.

```
JDialog dialog = new JDialog(this, "Attendance Reports & Defaulter Analysis", true);
```

Line Meaning (No Multiple Windows Bug): Yahan humne JFrame ke bajaye **modal JDialog** use kiya hai (jisme aakhri parameter true hai). Iska fayda ye hai ki jab report ya attendance khulti hai, toh naya alag window OS taskbar me spawn nahi hota, balki main window ke upar lock ho jata hai jab tak user use close na kare. Isse memory leak aur desktop clutter nahi hota.

```
// Student Portal Analytical Formula
int needed = (int) Math.ceil((3.0 * s.getTotalClasses() - 4.0 * s.getAttendedClasses()));
int canMiss = (int) Math.floor((4.0 * s.getAttendedClasses() / 3.0) - s.getTotalClasses());
```

Line Meaning (Attendance Advice Math):
• Agar student defaulter hai (< 75%), toh 75% target reach karne ke liye equation banti hai: (attended + x) / (total + x) >= 0.75. Isko cross multiply karke solve karne par aata hai: x = ceil(3*total - 4*attended). Itni upcoming classes usko lagatar attend karni padengi!
• Agar student eligible hai (>= 75%), toh wo kitni classes miss kar sakta hai: attended / (total + y) >= 0.75. Iska answer aata hai: y = floor((4*attended / 3) - total). Itne safe bunks wo afford kar sakta hai!

SECTION 4: "GHUMA KAR" PUCHHE JAANE WALE TRICKY VIVA QUESTIONS & CRISP ANSWERS

Ma'am direct sawal nahi puchhti, concept ko ghuma kar puchhti hain. In 8 questions ko ache se samajh lo:

Tricky Q1: "Tumne FileManager aur ValidationHelper ke methods ko 'static' kyu banaya? Kya ye OOPs ke khilaf nahi hai?"

Ans: "Nahi Ma'am, ye OOPs ke khilaf nahi balki standard Design Pattern (Utility Pattern) hai. FileManager aur ValidationHelper stateless utility classes hain — inka apna koi individual object state (instance variable) nahi hota. Agar hum static na banate, toh hume har baar bematlab 'new ValidationHelper()' object banana padta jo memory waste karta. Static banane se hum seedha 'ValidationHelper.isPositiveInteger()' call kar lete hain, jisse memory aur execution speed optimize hoti hai."

Tricky Q2: "CSV me tumne Attended aur Total classes dono kyu store kiye? Sirf final Percentage store kar lete toh file ka size chhota rehta na?"

Ans: "Ma'am, agar hum sirf percentage (jaise 85%) save karenge, toh agle din jab teacher attendance lagayega toh hum nayi percentage calculate nahi kar payenge! Kyunki hume ye nahi pata hoga ki kul kitni classes hui thi. Percentage ek derived attribute hai, isliye core counters (total aur attended) save karna mandatory hai taaki incremental math sahi chale."

Tricky Q3: "Table me checkbox lagane ke liye tumne kaun si external library use ki hai?"

Ans: "Ma'am, humne koi bhi third-party external library ya JAR file use nahi ki hai. Java Swing ke DefaultTableModel me ek method hota hai 'getColumnClass(int col)'. Humne us method ko override karke checkbox wale column ke liye 'Boolean.class' return kiya. Swing ka internal JTable engine internally boolean data type ko native interactive JCheckBox me automatically render kar deta hai."

Tricky Q4: "Agar user Roll Number me '101.5' ya 'abc' daal de toh tumhara code crash kyu nahi hota?"

Ans: "Ma'am, humne ValidationHelper me Integer.parseInt() ko structured try-catch block me wrap kiya hai jo specifically 'NumberFormatException' ko catch karta hai. Agar koi float number ya alphabet aata hai toh exception catch ho jaati hai, method false return karta hai, aur user ko clean modal error dialog dikhta hai without crashing the JVM."

Tricky Q5: "Pehle wale code me 60 dialogs ka problem tha, toh tumne usko solve kaise kiya technically?"

Ans: "Ma'am, Month 1 ke code me har student ke liye sequential JOptionPane.showConfirmDialog() loop me chal raha tha. Humne usko completely replace karke ek centralized JTable banaya. Ab teacher ek single screen par pure batch ke checkboxes dekhta hai, Select All dabata hai aur single Submit button click karte hi collection traverse hoke ek sath attendence mark ho jaati hai."

Tricky Q6: "Agar students.csv file delete ho jaye ya lab computer me missing ho toh kya application crash ho jayegi?"

Ans: "Nahi Ma'am, crash nahi hogi. Humare FileManager.loadStudents() me pehli line 'file.exists()' check karti hai. Agar file nahi milti toh 'createDefaultFile()' trigger hota hai jo turant data folder aur students.csv create karta hai aur 8 default student records insert kar deta hai. Isliye FileNotFoundException kabhi throw nahi hoti."

Tricky Q7: "Student aur AttendanceManager ke beech me kaun sa OOP relationship hai?"

Ans: "Ma'am, AttendanceManager aur Student ke beech me 'Aggregation' relationship hai (Has-A relationship), kyunki AttendanceManager ek ArrayList hold karta hai. Aur AttendanceManager aur FileManager ke beech me 'Dependency' relationship hai kyunki wo disk par read/write karne ke liye FileManager ke methods ko use karta hai."

Tricky Q8: "SwingUtilities.invokeLater() main method me kyu likha hai? Agar na likhein toh kya hoga?"

Ans: "Ma'am, Swing ke GUI components thread-safe nahi hote. SwingUtilities.invokeLater() ye ensure karta hai ki GUI creation aur event handling ka saara code background thread par nahi, balki Java ke dedicated Event Dispatch Thread (EDT) par chale, jisse UI hang nahi hoti aur thread race condition avoid hoti hai."

SECTION 5: STUDENT-WISE SPECIFIC VIVA QUESTIONS & READY ANSWERS

Har team member ko apne module par ye specific questions ready karke bolne hain:

1. Ashutosh Jitendra Singh (Data Persistence & FileManager.java)

Q: Tumne data save karne ke liye MySQL kyu nahi lagaya, CSV kyu chuna?

Ans: "Ma'am, college lab PCs par hume lightweight standalone tool chahiye tha. MySQL ke liye separate server install karna padta aur JDBC driver ka jar attach karna padta. CSV file standard flat-file format hai jo bina kisi external dependency ke chalta hai aur Microsoft Excel me directly open ho jata hai audit ke liye."

Q: PrintWriter me flush() method ka kya role hai?

Ans: "Ma'am, PrintWriter output ko seedha hard disk par nahi likhta balki pehle RAM buffer me rakhta hai. flush() call karne se buffer turant disk par empty ho jata hai jisse unexpected application close hone par data lose nahi hota."

2. Suman Kumar Gorain (GUI Layout & SmartAttendanceUI.java)

Q: Tumne JFrame ki jagah JDialog kyu use kiya reports aur batch marking me?

Ans: "Ma'am, jab hum new JFrame banate the toh har click par OS taskbar me ek nayi window khul jaati thi. Agar teacher ne 5 baar click kiya toh 5 windows ban jaati thi. JDialog modal hota hai, jo parent frame ke sath attached rehta hai aur memory leak hone se bachata hai."

Q: Table me bacho ka naam teacher galti se change na kar de, iska dhyan kaise rakha?

Ans: "Ma'am, DefaultTableModel me 'isCellEditable(int row, int col)' method override kiya hai aur usme sirf 'column == 4' (checkbox column) par true return karwaya hai. Baki sabhi columns read-only rehte hain."

3. Krish Kumar (Business Logic & Student Model)

Q: Constructor overloading ka real practical benefit kya hua is project me?

Ans: "Ma'am, naye admission ke waqt bache ki attendance 0 hoti hai toh hum 'Student(roll, name)' use karte hain. Lekin jab CSV file se purana attendance restore karte hain (jaise 20 attended out of 24), toh hume 4-argument wala constructor 'Student(roll, name, total, attended)' chahiye hota hai taaki purani history maintain rahe."

Q: Batch marking me total aur attended classes kaise calculate hoti hain?

Ans: "Ma'am, batch submit hote hi method ek loop me students ko traverse karta hai: har student ki totalClasses 1 badhti hai, aur jiska checkbox checked hota hai uski attendedClasses bhi 1 badh jaati hai, aur live percentage update ho jaati hai."

4. Preety Kumari (ValidationHelper.java & Security)

Q: Agar teacher roll number me negative number (-10) daal de toh kya hoga?

Ans: "Ma'am, hamare ValidationHelper me boundary check laga hai 'roll > 0'. Negative number aate hi method false return karta hai aur screen par warning aati hai ki 'Roll number must be a positive integer'."

Q: Duplicate roll number kaise prevent kiya?

Ans: "Ma'am, add student hone se pehle 'isRollDuplicate()' loop chala kar check karta hai ki candidate roll number pehle se list me kisi aur student ka toh nahi hai. Agar match milta hai toh dialog show karke admission rok diya jata hai."

5. Govind Kumar (System Architecture & Documentation)

Q: Is project ka 4-tier architecture explain karo.

Ans: "Ma'am, architecture 4 distinct layers me divide hai: 1) Model Layer (Student.java jo data hold karta hai), 2) Persistence Layer (FileManager.java jo CSV read/write karta hai), 3) Validation Layer (ValidationHelper.java jo input sanitize karta hai), aur 4) View Layer (SmartAttendanceUI.java jo screen aur table display karta hai). Aur AttendanceManager Controller ka kaam karta hai."

Q: UN-SDG Goal 4 se ye project kaise relate karta hai?

Ans: "Ma'am, SDG Goal 4 (Quality Education) ke Target 4.4 ke according ye application classroom proxy attendance ko eliminate karta hai aur digital automated reporting se institutional efficiency badhata hai."

SECTION 6: LIVE DEMO CHECKLIST (MAM KE SAMNE KYA KYA CLICK KARNA HAI)

Kal jab Mam project chala kar dekhne bolein, toh bina ghabraye in 5 steps me pura demo finish kar dena:

Step 1: Application Launch & Persistence Proof
Terminal me command run karo: java SmartAttendanceUI.
Mam ko batao: *"Ma'am dekhiye, background me FileManager ne turant 'data/students.csv' se 8 students load kar liye hain."*

Step 2: Teacher Login & Single-Screen Batch Marking
Click karo **Teacher Login** → Password daalo: admin123 → Faculty Dashboard khulega.
Click karo **1. Mark Daily Attendance (Batch Table)**.
Table khulegi. Mam ko dikhao: *"Ma'am yahan saare students ek screen par hain. Ab hum 'Select All' dabate hain, sab present ho gaye. Ek-do bacho ka tick hata kar Submit dabate hain."*
Pop-up aayega 'Attendance submitted and saved to CSV'. Mam ko dikhao ki ek click me sab save ho gaya!

Step 3: Defaulter Filter (< 75%) Show Karo
Dashboard me click karo **4. Defaulter List (< 75% Attendance)**.
Mam ko dikhao: *"Ma'am dekhiye, Priya Singh (66.67%), Preety Kumari (58.33%), aur Rohan Verma (50.00%) automatically filter ho gaye aur unke aage Red me 'Critical / Defaulter' status aa raha hai. Aur upar 'All Students' dabane par pura batch wapas aa jata hai."*

Step 4: Input Validation & Duplicate Protection Show Karo
Click karo **2. Add New Student**.
Roll number me daalo -5 ya 101 (jo already Aarav ka roll hai) → Error pop-up aayega.
Mam ko batao: *"Ma'am, ValidationHelper layer invalid inputs aur duplicate roll numbers ko intercept kar leti hai bina crash hue."*

Step 5: Student Portal Attendance Card & Mathematical Advice
Main menu me aao aur click karo **Student Portal**.
Roll number daalo 105 (Ashutosh) → Green me **ELIGIBLE FOR EXAM (100%)** aayega aur advice aayegi ki kitne classes safe zone me bunk kar sakte hain.
Phir Roll number daalo 102 (Priya) → Red me **CRITICAL / SHORT ATTENDANCE (66.67%)** aayega aur mathematical advice aayegi: *'Need to attend next X consecutive classes to reach 75%.'*